

Breast Cancer Assessment

University of Bristol x RUH Bath

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Dr Nicola Laurence

Project Introduction



Our Client(s)

**Working with breast surgeon Dr Nicola Laurence, at RUH
Bath Hospital.**

**Over time, we have collaborated with several other
doctors from different hospitals in the South West and
research groups.**



What have we created?

Desktop application for doctors/surgeons.

They can input patient oncology data and personal details which is used to calculate and return a list of eligible treatments for the patient. Retains a table of patients.

Information is all stored locally encrypted.



Why?

**Speed up the patient treatment decision-making process
to reduce waiting times**

**Improve data handling and help ease the pressure off
doctors/surgeons, allow flexible working across
departments.**

**Previous treatment decision-making process was through
group meetings called “MDTs”**



User Story

At meetings, Dr Nicola Lawrence can easily enter the patient data into her application and view the eligible treatments.

- **Eliminates the need for mental calculations of treatments**
- **Minimises risk of human error and ensures adherence to current treatment criteria - and new ones!**
- **Family history can also be calculated as opposed to organising multiple teams at the hospital**



Demonstration



What we were given

- A list of parameters split into 4 categories
- A list of 'Algorithms' using the given parameters written in simple English. i.e. "Age Under 50 and HER2 is 1+ or 2+ and cN N2 or N3"
- Some inconsistent colour coding
- Continuous updates

	AA	AB	AC	AD	
1	oncotype	CDK 4/6 I	CPS + EG score	Mainstream testing R444/ PARP	adjuvant Kadcyla
2	eligible	eligible		eligible	eligible
3	not eligible	not eligible		not eligible	not eligible
4					
5					
6					
7					
8					
9	Nx or N0 or N1(mic) or N1		clinical Preop TNM		
10	and if less than 80 yrs	ER (4 or 5 or 6 or 7 or 8)		Her 2 0 or 1+ or 2+ and	her 2 3+
11	and ER 6, 7 or 8	and Her 2 (0 or 1+ or 2+)	I: (cT1 or Tx) and N0 =0	neoadjuvant SCAT and	and neoadjuvant S
12			IIA: (cTx and cN1), or (cT1 and cN1) or (cT2 and cN0) or (cTis and cN1)=0		
13	and Her 2 0 or 1+ or 2+	and (N2) or (pN1 and postop grade 3) or (pT3 and pN1)	IIB: (cT2 and N1) or (cT3 and N0)=1	ER 0 or 1 or 2 or 3 and	and all TNM exce
14	and pT2 and postop grade 2 or any pT and postop grade 3		IIIA: ((cTx or 1 or T2) and N2) or (cT3 and cN1) or (cTis and cN2)=1	all pTN except pT0N0	
15	and no neoadjuvant SCAT		IIIB: cT4 and (N0 or N1 or N2)=2	OR	
16		just checking that ER 4 is in here as it was a late change	IIIC: (cTx or cTis or cT1 or cT2 or cT3 or cT4) and N3 =2	Her 2 0 or 1+ or 2+ and	
17				neoadjuvant SCAT and	
18	ADD IN N1 (mic) if not already in			ER 4 or 5 or 6 or 7 or 8 and	
19				CPS + EG score > or = 3	
20				OR	
21			postop TNM	or her 2 (0 or 1+ or, 2+) and	
22				no neoadjuvant SCAT	
23			0: pT0 or Tis	and ER 0 or 1 or 2 or 3	
24			I: (pT1 and N0) or (pT0 and N1(mic)) or (pT1 and N1(mic)) or (pTis and pN1) and pN1 or pT2 or pT3 or pT4		
25			IIA: (pTx and pN1), or (pT1 and pN1) or (pT2 and pN0) =1	OR	
26			IIB: (pT2 and pN1) or (pT3 and pN0) =1	or Her 2 (0 or 1+ or 2+) and	
27			IIIA: ((pTx or T1 or T2) and pN2)) or (pT3 and pN1) or (pTis and pN2) =1	no neoadjuvant SCAT	
28			IIIB: pT4 and (pN0 or pN1 or pN2)=1	and ER 4 or 5 or 6 or 7 or 8	
29			IIIC: (pTx or or pTis or pT1 or pT2 or pT3 or pT4) and pN3 =2	and pN2 or pN3	
30					
31					
32			INCLUDE ALL pT1 (ie Tmic, T1a T1b etc) and all pT4		
33					
34			post op grade 1 or 2 = 0	INCLUDE ALL T4	
35			post op grade 3 = 1		
36					
37			TOTAL SCORE to feed into next column CPS + EG score		
38					
39					
40			IF TOO DIFFICULT TO ADD IN Tis then it is OK TO LEAVE OUT, hopefully all T1 and T4 are in		
41					
42					

<

>

...

postop information (tumour 1)

postop information (tumour 2)

Treatment given

+

:

◀

▶

	A	B	C	D	E	F	G	H	I	J	K	
1	neoadjuvant SCAT	Neoadjuvant chemotherapy	Neoadjuvant anti her 2 treatment	Neoadjuvant immunotherapy	Neoadjuvant endocrine (16 weeks)	previous Radiotherapy on same side	Surgery	Axilla	Postop Size (mm)	postop grade	post op Ki67 %	Closest ra
2	yes	yes with anthracycline	yes herceptin	yes		yes	Breast conservation	none	number	LG DCIS		
3	no	Yes, without anthracycline	yes phesgo	no		no	Mastectomy	sample		IG DCIS		
4		yes , unknown	yes other					SNB		HG DCIS		
5		no	no					ANC		Invasive grade 1		
6										Invasive grade 2		

[illegible]

breast cancer with patient and/or female relative under 30 on mothers side	breast cancer with patient and/or female relative 30-39 on mothers side	breast cancer with patient and or female relative 40-49 on mothers side	breast cancer with patient and/ or any female relative 50-59 on mothers side	breast cancer with patinet and/or any female relative>59 on mothers side	male breast cancer <60 on mothers side	male breast cancer >59 on mothers side	ovarian cancer on mother's side
0	0	0	0	0	0	0	
1	1	1	1	1	1	1	
2	2	2	2	2	2	2	
etc	etc	etc	etc	etc	etc	etc	
score 11 for each relative <i>manchester score</i>	score 8 for each relative <i>manchester score</i>	score 6 for each relative <i>manchester score</i>	score 4 for each relative <i>manchester score</i>	score 2 for each relative <i>manchester score</i>	score 13 for each relative <i>manchester score</i>	score 10 for each relative <i>manchester score</i>	score 1 for each relative <i>manchester score</i>

P	Q	R	S	T	U	V
RT to chest wall	RT to axilla and SCF	RT to SCF alone	Aromatase inhibitor	Tamoxifen	bisphosphonate	oncotype
eligible	eligible	eligible	eligible	eligible	eligible	eligible
not eligible	not eligible	not eligible	not eligible	not eligible	not eligible	not eligible
						Nx or N0 or N1(mic) or N1
mastectomy			if post menopausal and		if postmenopausal	and if less than 80 yrs
and No previous RT and		ANC	ER 4 or 5 or 6 or 7 or 8	ER 4 or 5 or 6 or 7 or 8	and grade 2 or 3	and ER 6, 7 or 8
pT3 or pT4	SNB or sample	and pN2 or pN3 or (cN2 or cN3 and neoadjuvant SCAT)				and Her 2 0 or 1+ or 2+
or pN1 or pN2 or pN3 (not N1(mic))	and cN1 or pN1 (not N1 (m	and No previous RT				and pT2 and postop grade 2 or any pT and postop grade 3
or margins positive with invasive disease ie						
closest radial margin 0mm						
	and No previous RT					and no neoadjuvant SCAT
Or cT4 (any) and neoadjuvant scat						
		ADD IN				ADD IN N1 (mic) if not already in
ADD IN						
breast conservation surgery	breast conservation surgery					
and over 50 years old	and over 70 years old	mastectomy				if post menopausal and
and invasive ductal ca	and pT1	and No previous RT and			ANC	ER 4 or 5 or 6 or 7 or 8
and Nx or N0	and Nx or N0					
and pT1	and postop Grade 1 or 2	pT3 or pT4		SNB or sample	and pN2 or pN3 or (cN2 or cN3 and neoadjuvant SCAT)	
and postop Grade 1 or 2	and ER 4 or 5 or 6 or 7 or 8	or pN1 or pN2 or pN3 (not N1(mic))		and cN1 or pN1 (not N1 (m	and No previous RT	
		or margins positive with invasive disease ie				
and ER 4 or 5 or 6 or 7 or 8		closest radial margin 0mm				
	and HER2 1+ or 2+ or 0			and No previous RT		
and HER2 1+ or 2+ or 0	and No previous RT	Or cT4 (any) and neoadjuvant scat				
and No previous RT	and no neoadjuvant treatment					
and no neoadjuvant treatment (A-E)	and not mulitfocal				ADD IN	
and not mulitfocal						

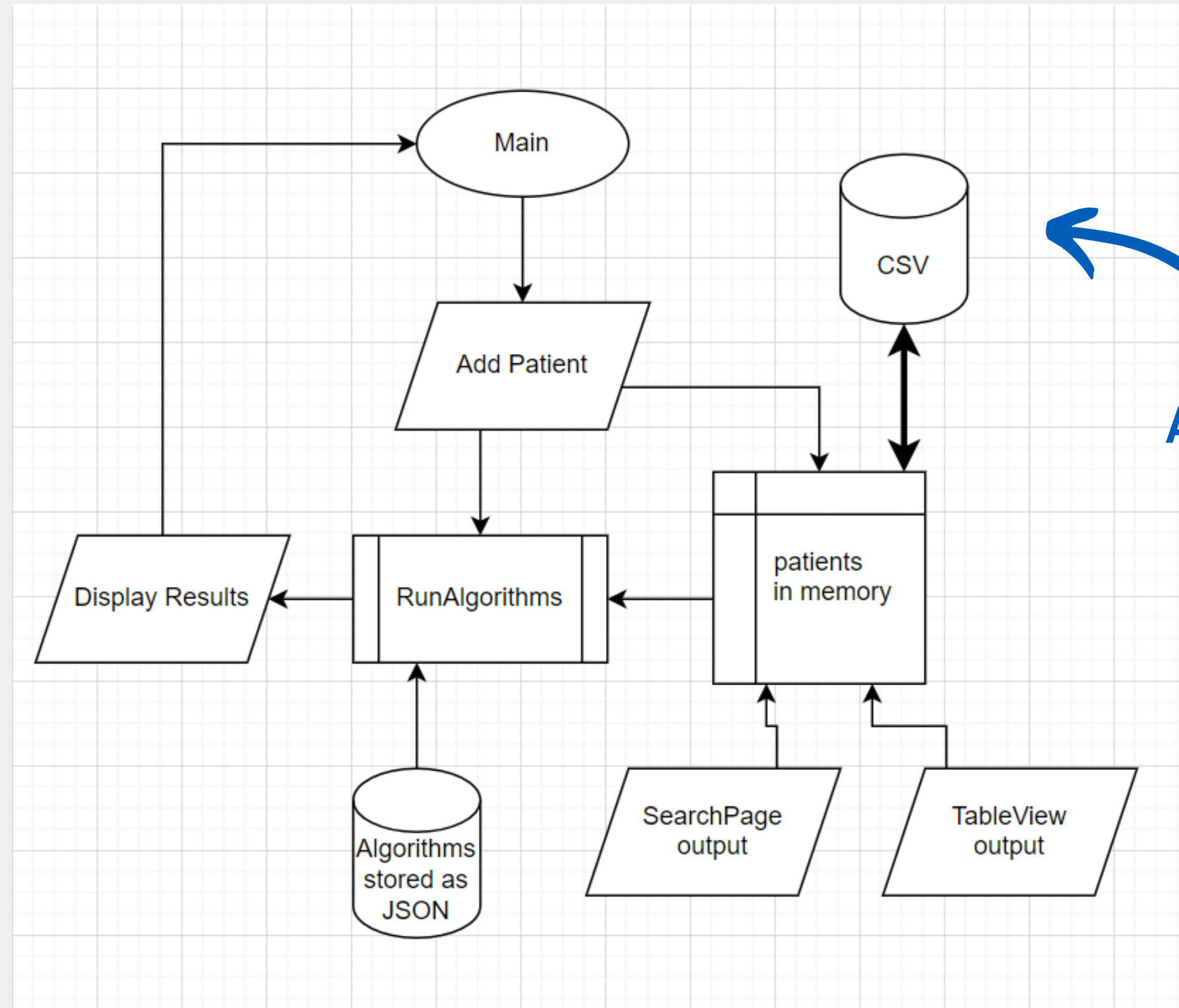
P
Clinical Stage
I
IIA
IIB
IIIA
IIIB
IIIC
I: (cT1 or Tx) and N0
IIA: (cTx and cN1), or (cT1 and cN1) or (cT2 and cN0) or (Tis and N1)
IIB: (cT2 and N1) or (cT3 and N0)
IIIA: ((cTx or 1 or T2) and N2) or (cT3 and cN1) or (Tis and N2)
IIIB: cT4 and (N0 or N1 or N2)
IIIC: (cTx or cTis or cT1 or cT2 or cT3 or cT4) and N3
INCLUDE ALL T1 (ie a,b and c)and all T4 (ie a, b, c and d)

W	X	Y
CDK 4/6 I	CPS + EG score	Mainstream testing R444/ PARP
eligible		eligible
not eligible		not eligible
	clinical Preop TNM	
ER (4 or 5 or 6 or 7 or 8)		Her 2 0 or 1+ or 2+ and
and Her 2 (0 or 1+ or 2+)	I: (cT1 or Tx) and N0 = 0	neoadjuvant SCAT and
	IIA: (cTx and cN1), or (cT1 and cN1) or (cT2 and cN0) or (Tis and N1)= 0	
	IIB: (cT2 and N1) or (cT3 and N0) =1	ER 0 or 1 or 2 or 3 and
and (N2) or (pN1 and postop grade 3) or (pT3 and pN1)	IIIA: ((cTx or 1 or T2) and N2) or (cT3 and cN1) or (Tis and N2)=1	all pTN except pT0N0
	IIIB: cT4 and (N0 or N1 or N2)=2	OR
just checking that ER 4 is in here as it was a late change	IIIC: (cTx or cTis or cT1 or cT2 or cT3 or cT4) and N3 =2	Her 2 0 or 1+ or 2+ and
		neoadjuvant SCAT and
		ER 4 or 5 or 6 or 7 or 8 and
		CPS + EG score > or = 3
		OR
	postop TNM	or her 2 (0 or 1+or ,2+) and
		no neoadjuvant SCAT
	0: pT0 or Tis	and ER 0 or 1 or 2 or 3
	I: (pT1 and N0) or (pT0 and N1(mic)) or (pT1 and N1(mic)) or (pTis and N1)=0	and pN1 or pT2 or pT3 or pT4
	IIA: (pTx and pN1), or (pT1 and pN1) or (pT2 and pN0) =1	OR
	IIB: (pT2 and pN1) or (pT3 and pN0) =1	or Her 2 (0 or 1+ or 2+)and
	IIIA: ((pTx or T1 or T2) and pN2)) or (pT3 and pN1) or (pTis and pN2) =1	no neoadjuvant SCAT
	IIIB: pT4 and (pN0 or pN1 or pN2)=1	and ER 4 or 5 or 6 or 7 or 8
	IIIC: (pTx or or pTis or pT1 or pT2 or pT3 or pT4) and pN3 =2	and pN2 or pN3
	INCLUDE ALL pT1 (ie Tmic, T1a T1b etc) and all pT4	
	post op grade 1 or 2 = 0	
	post op grade 3 = 1	
	TOTAL SCORE to feed into next column CPS + EG score	
	IF TOO DIFFICULT TO ADD IN Tis then it is OK TO LEAVE OUT, hopefully all T1 and T4 are in	

high risk if (MATERNAL SIDE) INCLUDE PATIENT AS FIRST DEGREE RELATIVE	high risk if PATERNAL SIDE INCLUDE PATIENT AS FIRST DEGREE RELATIVE
yes as per maternal side	yes as per paternal side
no	no
2 x first degree relative with breast cancer less than an average age of 50	2 x first degree relative with breast cancer less than an average age of 50
or 1 x first degree relative and 1 x second degree relative with breast cancer less than average age of 50 (combine ages of 2 people and calculate mean)	1 x first degree relative and 1 x second degree relative with breast cancer less than average age of 50
or 3 x first degree relatives with breast cancer less than an average age of 60 (combine the ages of 3 people and calculate mean)	3 x first degree relatives with breast cancer less than an average age of 60
or 2x first degree relative and 1 x second degree relative with breast cancer less than average age of 60	2 x first degree relative and 1 x second degree relative with breast cancer less than average age of 60
or 1x first degree relative and 2 x second degree relative with breast cancer less than average age of 60	1 x first degree relative and 2 x second degree relative with breast cancer less than average age of 60
or 1x first degree relative and 3 x second degree relative with breast cancer less than average age of 60	1 x first degree relative and 3 x second degree relative with breast cancer less than average age of 60
or 2x first degree relative and 2 x second degree relative with breast cancer less than average age of 60	2 x first degree relative and 2 x second degree relative with breast cancer less than average age of 60
or 3x first degree relative and 1 x second degree relative with breast cancer less than average age of 60	3 x first degree relative and 1 x second degree relative with breast cancer less than average age of 60
or 1 x ovarian cancer and 1 x first degree relative	1 x first degree relative
or 1 x ovarian cancer and 1 x second degree relative under 50	1 x second degree relative under 50
or 1 x ovarian cancer and 1 x first degree relative and 1 x second degree relative with breast cancer less than average age of 60	1 x first degree relative and 1 x second degree relative with breast cancer less than average age of 60
or more than one relative with ovarian cancer	more than one relative with ovarian cancer
or 1 x first degree relative with breast cancer on both sides, both diagnosed under the age of 50.	1 x first degree relative with breast cancer on both sides, both diagnosed under the age of 50.
or 1 x first degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.	1 x first degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.
or 1 x second degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.(the second degree relative can be any age)	1 x second degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.
or 1 x second degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.	1 x second degree relative with breast cancer on both sides and 1 x first degree relative breast cancer under 60.
or 1 x first degree relative with breast cancer on both sides and 1 x second degree relative breast cancer under 60.	1 x first degree relative with breast cancer on both sides and 1 x second degree relative breast cancer under 60.
or 1 x male breast cancer and 1 x first degree relative under 50	1 x male breast cancer and 1 x first degree relative under 50
or 1 x male breast cancer and 1 x second degree relative under 50	1 x male breast cancer and 1 x second degree relative under 50
or 1 x male breast cancer and 2 x first degree relative under average age 60	1 x male breast cancer and 2 x first degree relative under average age 60
or 1 x male breast cancer and 1 x first degree relative and 1 x second degree relative under average age 60 (average the age of the first and second degree relatives)	1 x male breast cancer and 1 x first degree relative and 1 x second degree relative under average age 60
or 1 x male breast cancer and 2 x second degree relative under average age 60	1 x male breast cancer and 2 x second degree relative under average age 60
or 1 x first degree relative and ashkenazi jewish hertiage	1 x first degree relative and ashkenazi jewish hertiage

And sooooo
many more...

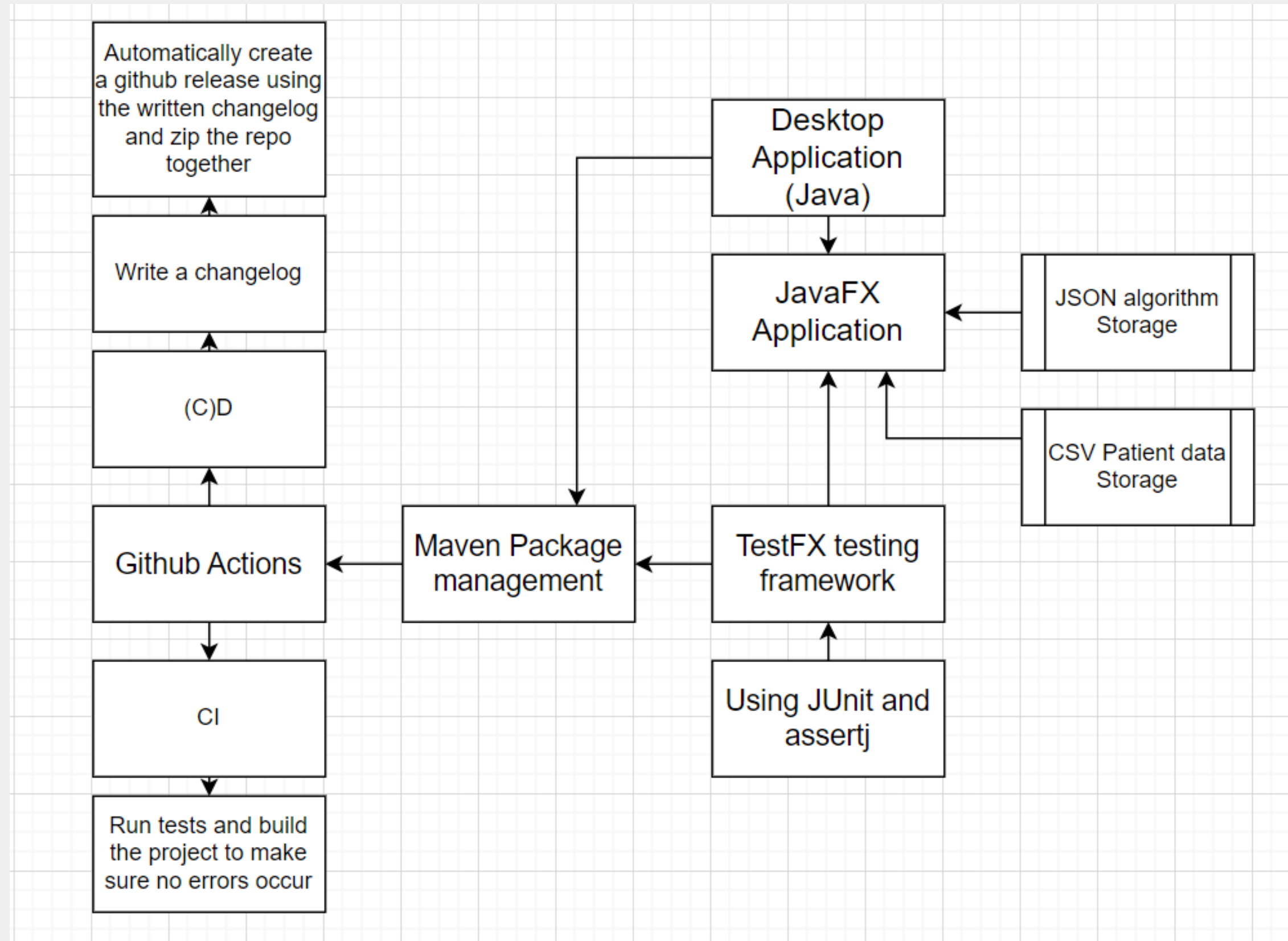
Implementation: Data Flow



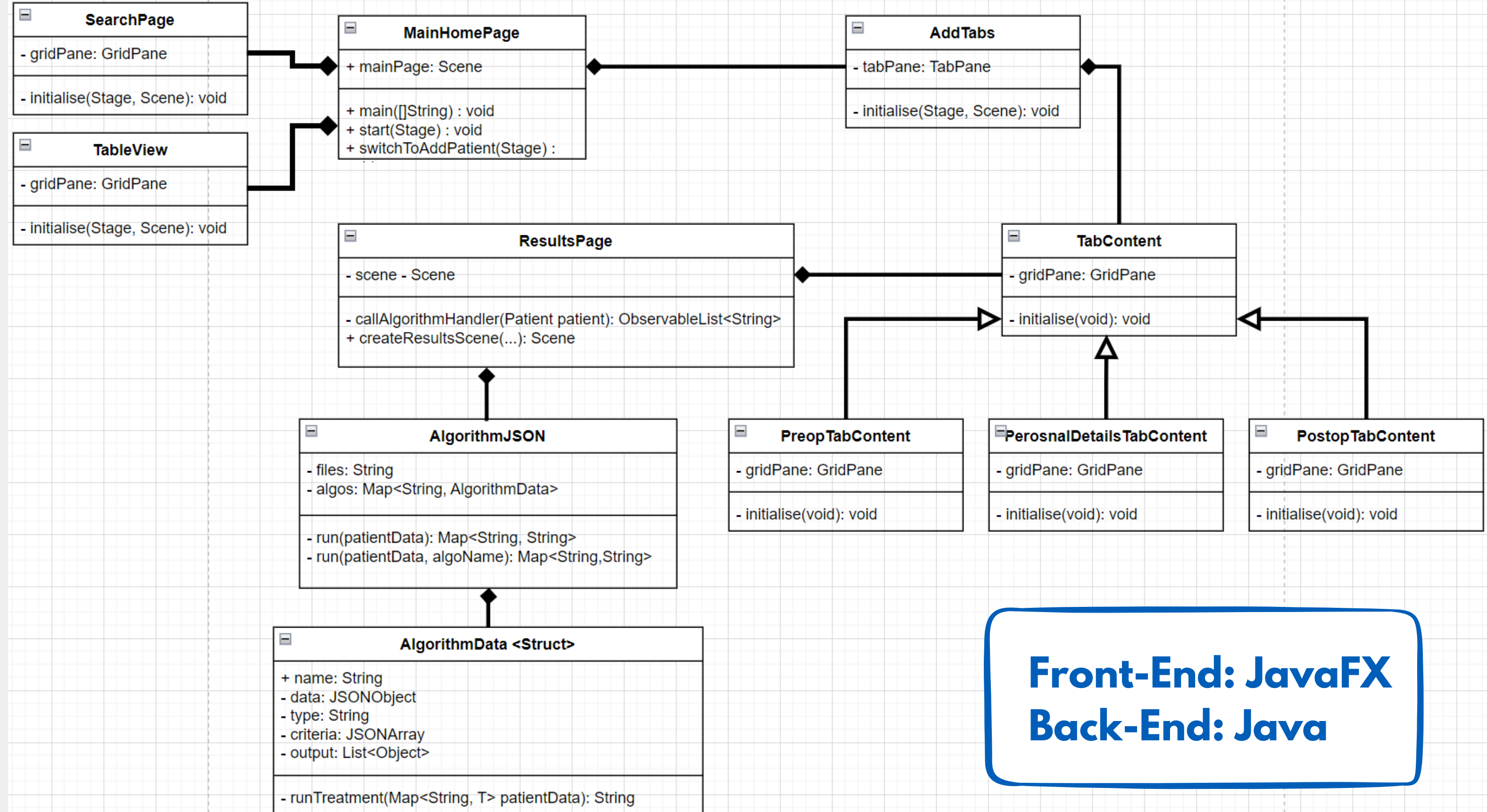
AES Encrypted JSON



Architecture + Workflow



Our Tech Stack



Front-End: JavaFX
Back-End: Java

Challenges faced

1

Lost a team member initially... :(
...but then gaining a new team member: Lamb! :D
...then lost yet another team member... :(

Onboarded Lamb with our goals and current progress and
organised new meetings with our client!



Challenges faced

Technical

2

Setting up each our individual laptops
(the Curse of JavaFX and VM Options)

Perfected our gitignore files! (And a file fallback
mechanism).



Challenges faced

3

Client constantly changing the algorithms and parameters after sending us the file

Purposely delayed our progress to ensure we only used the most accurate information.

“17”



Parameters.and
algoriththms
version 17_ (1)

Challenges faced

Technical

4

Understanding, then designing the algorithms and calculations using that complex file...

Especially the Family History Section

A very difficult logic puzzle of and's and ors and combos of them!



Challenges faced

5

Waiting for NHS approval on database implementations

Finally given “Permission” 2/3 weeks before final release



Challenges faced

Technical

6 NHS Data Struggle

Read Information Governance document given to us.
Implemented a rudimentary data encryption as proof of concept.



Challenges faced

Technical

7

A whole range of issues, from syntax formatting to ensuring **robust** data encryption & writing **JSON** algorithms.

```
    "Oncotype" : {  
      "type" : "treatment",  
      "criteria" : [  
        [{"PN" : ["NX", "N0", "N1", "N1_MIC"]} ],  
        [{"AGE_UNDER": ["80"]} ],  
        [{"ER" : [6,7,8]} ],  
        [{"HER2" : ["ZERO", "PLUS_1", "PLUS_2"]} ],  
        [{"NEO_ADJ_SACT" : ["NO"]} ],  
        [  
          {"PT" : ["T2"], "POST_OP_GRADE" : ["INVASIVE_GRADE_2"]},  
          {"PT" : ["T0", "T1"], "POST_OP_GRADE" : ["INVASIVE_GRADE_3"]}  
        ]  
      ],  
      "output" : ["eligible", "not eligible"]  
    },  
  ],  
}
```

